

The empirical recurrence for OEIS A251354: recovering a conjecture that is not written down

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Abstract

OEIS A251354 records “Empirical recurrence of order 96”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on $S = 256$ vertices and satisfies some monic recurrence of order at most S ; Berlekamp–Massey applied to $2S$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 96, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 96 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A251354 is “Number of $(n+1) \times (3+1)$ 0..3 arrays with no 2×2 subblock having its maximum diagonal element less than the absolute difference of its antidiagonal elements.”. It has offset 1 and begins

39594, 6505867, 1069722970, 175856353691, 28911058486029, 4753101689364667, 781429834781394639, 12847026

The entry states:

Empirical recurrence of order 96 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 13:17:26, revision 6) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 256$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S = 256$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 512$ exact terms of the model returns order 96 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{96} c_i a(n-i),$$

c_1	256	c_{25}	−124928738953008686340968596	c_{49}	196550294922438255
c_2	−20306	c_{26}	1128172055547175595684911733	c_{50}	−262472190892650968
c_3	1096141	c_{27}	−7178938552563705187353868462	c_{51}	331021841650514348
c_4	−46130653	c_{28}	35292956425953488554158557489	c_{52}	−389019714013797738
c_5	1465746894	c_{29}	−141945824219498544242257611321	c_{53}	420831003197763563
c_6	−35293177101	c_{30}	477417253185896531052585045744	c_{54}	−413743843764710374
c_7	669656498716	c_{31}	−1326054635716980262299866678213	c_{55}	363553489046683960
c_8	−10439713264260	c_{32}	2807521203275483091466039082948	c_{56}	−277461860121756093
c_9	137839981216810	c_{33}	−2893765500662719343527801000213	c_{57}	172421375311319601
c_{10}	−1569935153952007	c_{34}	−10509805256004341131708073305653	c_{58}	−69417429639591125
c_{11}	15540271468945659	c_{35}	84463373805260752688434177937503	c_{59}	−13617633282454113
c_{12}	−132853833549530938	c_{36}	−366758586427219064031975572525920	c_{60}	669748576205596055
c_{13}	958418417606589452	c_{37}	1254707686655723870359224455381533	c_{61}	−90415660043309319
c_{14}	−5564358137451893092	c_{38}	−3682647913914106174927037485510530	c_{62}	907085306892385055
c_{15}	23032026782442858326	c_{39}	9582782633418438954302472772434736	c_{63}	−77409524689723914
c_{16}	−31455493446232585383	c_{40}	−22451603160518752838831199278045978	c_{64}	591375660389595596
c_{17}	−518984897073177330293	c_{41}	47769110218196978200600495000582604	c_{65}	−41618993729559830
c_{18}	6318490397360266499044	c_{42}	−92916949884731655066792416260628708	c_{66}	275179264604548187
c_{19}	−44762364025609539817414	c_{43}	166549572444435287699358309775803181	c_{67}	−17319800678111735
c_{20}	232248525204359936223354	c_{44}	−278169389825801345581879902318729781	c_{68}	104379579235646726
c_{21}	−904517875995871040727500	c_{45}	439204186207743768242558484944757110	c_{69}	−60139704335497339
c_{22}	2531666362839233699341104	c_{46}	−665855485260863474153695377757903154	c_{70}	328985311045450830
c_{23}	−4798726334370278005843550	c_{47}	981039869522012932379017453341622458	c_{71}	−16930347418276622
c_{24}	12360750534002113385359257	c_{48}	−1410021510294587603945555810525800547	c_{72}	81229793739899683

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 96, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $96 + 4$ terms removed returns order 96 as well, so $\deg q_{\min} = 96 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 11 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $96 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 96 that this sequence satisfies — and that family turns out to have exactly one member.

References

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